# Attention and transformers

## AI · University · Lecture 3



# In this lecture

- Compute scaled dot-product attention.
- Explain $\sqrt{d}$ and position.
- Sequence the transformer for novices.

# Remember

- Attention is a weighted average.
- ÷√d prevents saturation.
- One head by hand first.

# One head by hand

1. Warm-up (5 min): Which noun?
2. Compute (20 min): A 3-token head.
3. Break (10 min): Remove √d.
4. Build (20 min): Add the parts.
5. Wrap-up (5 min): One-minute paper.

# Let's discuss

1. What happened to the weights without √d?
2. Can attention maps explain a prediction?
3. Which part will you teach last?

# Exit ticket

- Write $\text{softmax}(QK^T/\sqrt{d})V$ and label what each letter means.